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Analyst Lab Africa

Experience Lab Internship

SwiftLog Logistics Analytics & AI Transformation: Executive Business Intelligence Report

Data Analytics Track

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Week 5: EXECUTIVE BUSINESS INTELLIGENCE REPORT

1. Project Overview

SwiftLog Logistics Limited is a fast-growing Nigerian logistics and supply chain company. Management has reported late deliveries, pressure on inventory, uneven supplier performance and little visibility of key numbers. This report summarises the Power BI solution built for SwiftLog in Week 5 of the Experience Lab Internship. The solution has five report pages: Performance Overview, Customers & Order Analytics, Inventory & Warehouse Analytics, Shipment & Delivery Analytics, and Supplier & Product Performance Analytics.

The report gives the main KPIs, what each page shows, answers to the ten business questions in the brief, the items flagged during review, and ten recommendations. Where a number was read from a chart instead of a card or table, it is marked with "~" and should be treated as approximate. Power Query steps, the data model and the DAX measure documentation are covered in the accompanying project files.

2. Part A: Data and Dashboard Review

Scope. The model combines the SwiftLog customer, order, order item, product, supplier, warehouse, inventory, shipment, delivery performance, employee and return datasets. It covers 5,000 orders from 1,000 customers, 5,000 shipments, 250 products, 150 suppliers and 15 warehouses. Orders run from January to September.

Consistency checks. Because the five pages draw on one model, their figures should agree. The checks below were run on the numbers shown in the dashboards. Everything agreed except three items, which are listed in Part E (F6, F7 and F10).

Check	Result
Total revenue	₦21.67M on the Overview, ₦21,668,065.85 as the Customers table total, and ₦21.67M shipment value on the Shipments page. Agree.
Order count	5,000 in the status chart (1,716 pending + 1,658 delivered + 1,626 cancelled), in the customer table total and in the warehouse table total. Agree.
Average order value	₦21,668,066 ÷ 5,000 = ₦4,334. The card shows 4.33K. Agree.
Revenue per supplier	₦21,668,066 ÷ 150 = ₦144,454. The card shows 144.45K. Agree.
Employees per warehouse	200 employees (108 Lagos + 54 Abuja + 38 Kano) ÷ 15 warehouses = 13.33. Agree.
On-time delivery rate	36.28% on both the Overview and the Shipments page. Agree.
Delay days	3.57 average × 5,000 shipments = 17,850 delay days. The status chart shows ~16K delayed plus ~1.8K on time. Agree.
Category revenue	Fashion ~₦8.1M + Food ~₦7.1M + Electronics ~₦6.5M = ~₦21.7M. Agree.

KPI definitions as built. Total Revenue is the value of all orders whatever their status. On-Time Delivery Rate is the share of shipments with status On Time. Average Delay Days is total delay days divided by all shipments. Cancellation Rate is cancelled orders divided by all orders. Low Stock Items is the count of items flagged as low stock; the threshold is not shown on the dashboard.

3. Part B: KPI Summary

KPI	Value	Comment	Status
Total Revenue	₦21.67M	Includes cancelled and pending orders (F1)	Watch
Total Orders	5,000	About 556 per month from January to August	Normal
Total Customers	1,000	5.00 orders per customer	Normal
Total Products / Suppliers	250 / 150	₦144.45K revenue per supplier	Normal
Total Shipments	5,000	Equal to order count, including cancelled orders (F7)	Watch
Average Order Value	₦4,334	Shown as 4.33K	Normal
Inventory Stock Level	1,007,764 units	380 low stock items; threshold not stated (F9)	Watch
On-Time Delivery Rate	36.28%	About 3,186 shipments (63.72%) were late	Attention
Average Delay	3.57 days	Taken across all shipments, so late shipments average more	Attention
Cancellation Rate	32.52%	1,626 cancelled orders	Attention
Pending Orders	1,716	34.32% of all orders	Attention
Return Rate	9.46%	No product-level return rate shown yet	Watch
Employees per Warehouse	13.33	200 staff across 15 warehouses	Normal

Status is the author's judgement: Attention means management action is needed now, Watch means the figure needs a definition or a check, Normal means no concern from the dashboard.

4. Part C: Dashboard Findings

Performance Overview. Monthly revenue held between about ₦2.5M and ₦2.85M from January to August and peaked in August. September drops to about ₦0.6M, which looks like a partial month and not a real fall (F5). Fashion leads on category revenue at ~₦8.1M (37%), ahead of Food at ~₦7.1M (33%) and Electronics at ~₦6.5M (30%). The two weakest cards are on-time delivery (36.28%) and return rate (9.46%).

Customers & Order Analytics. The 5,000 orders split almost evenly by status: 34.32% pending, 33.16% delivered and 32.52% cancelled, so only one order in three is delivered. Kano, Lagos, Ibadan and Abuja each supply about a quarter of orders and revenue, with Kano highest at ~1,280 orders and ~₦5.7M. In the scatter of total orders against cancelled orders, cancellations (up to 6 per customer) appear across the whole customer base, which points to a process problem more than to a few customers.

Inventory & Warehouse Analytics. Stock stands at 1,007,764 units, with 380 low stock items. Lagos holds ~54% of the stock and 54% of the staff (108 of 200), Abuja ~27% and 27% (54), and Kano ~19% and 19% (38), so staffing follows stock closely. By category, Fashion holds ~0.38M units, Electronics ~0.325M and Food ~0.305M. Among the ten warehouses visible, order volume is level, from 319 orders (W06) to 371 (W07).

Shipment & Delivery Analytics. Only 36.28% of shipments arrive on time. About 3,186 are late and the average delay is 3.57 days. Lagos accounts for ~9.5K delay days, Abuja ~4.7K and Kano ~3.6K. Shipments follow the revenue pattern: ~590 in January, a peak of ~655 in August and ~150 in September. Delay days and returns rise together across locations (Lagos ~250 returns, Abuja ~140, Kano ~110), but with only three points this does not show cause.

Supplier & Product Performance Analytics. Revenue per supplier is ₦144.45K. Supplier 40 (~₦0.32M), Supplier 6 (~₦0.31M) and Supplier 70 (~₦0.30M) lead, each earning more than twice the average. Monthly revenue per supplier is steady at ~₦17K to ₦19K until the September drop. The page measures revenue only; it has no delivery or return measure for suppliers yet.

5. Part D: Business Analysis

1. Which products generate the highest revenue? By category, Fashion (~₦8.1M, 37%), then Food (~₦7.1M) and Electronics (~₦6.5M). The current visuals do not rank single products, so a top-10 product chart is still needed.

2. Which customers contribute the highest revenue? Average revenue is ₦21,668 per customer. Of the ten customers shown, Customer 4 (₦42,382) and Customer 2 (₦31,667) are highest, about twice and one and a half times the average. The table is sorted by customer ID, so a ranked list is still needed.

3. Which warehouses operate most efficiently? Order load is level (319 to 371 orders across W01 to W10), so no warehouse carries an unusual load. If staff are spread evenly, Lagos runs about eight warehouses, Abuja four and Kano three, and each then records roughly 1,200 delay days. On that reading no site stands out as better or worse. A delay rate for each warehouse would confirm this.

4. Which suppliers consistently perform well? On revenue, Suppliers 40, 6, 70, 85 and 100 lead. Revenue is not reliability, and delivery performance by supplier is not on the dashboard, so this report cannot yet say which suppliers are dependable.

5. Which products have the highest return rates? The overall rate is 9.46%. By location, Lagos has the most returns (~250), then Abuja (~140) and Kano (~110). No product or category return rate is shown yet.

6. Which regions experience the most delayed deliveries? Lagos, with ~9.5K delay days (about 53% of the total), then Abuja (~4.7K) and Kano (~3.6K). This matches each location's share of warehouses (see question 3), so the delay problem looks network-wide.

7. Which months record peak business activity? August is the peak (~₦2.85M revenue, ~655 shipments), followed by April (~₦2.75M, ~640 shipments). September is not comparable because it looks incomplete.

8. Which KPIs need immediate management attention? On-time delivery (36.28%), cancellation rate (32.52%), pending orders (1,716), average delay (3.57 days), return rate (9.46%) and the 380 low stock items.

9. What operational inefficiencies were identified? Late delivery is the norm (about 64% of shipments). A third of orders are cancelled and a third are still pending. Revenue is reported before cancellations are removed. Lagos holds ~54% of stock but Lagos customers place only ~25% of orders, and Ibadan places ~1,240 orders with no Ibadan warehouse in the data. Whether this stock position adds to delays should be tested.

10. What business opportunities should SwiftLog pursue? Recover cancelled orders (about ₦7.0M of order value if cancelled orders carry the average value), raise the on-time rate, build on Fashion, move stock closer to Kano, Abuja and Ibadan demand, and score suppliers so that volume goes to the dependable ones.

6. Part E: Flagged Items

The items below were found while reviewing the five pages. Priority reflects the effect on decisions: Critical and High items should be dealt with first.

Ref	Page	What was found	Action	Priority
F1	Overview, Customers	Revenue is not filtered by order status. The ₦21.67M total equals the value of all 5,000 orders. If cancelled and pending orders carry average value, about ₦7.0M is cancelled, ₦7.4M is pending and ₦7.2M sits on delivered orders.	Report delivered-order revenue as the main KPI. Keep booked revenue as context.	High
F2	Overview, Shipments	On-time delivery is 36.28%. About 3,186 shipments were late.	Set a target and review late shipments weekly.	Critical
F3	Customers	Cancellation rate is 32.52% (1,626 orders). No cancellation reason is shown.	Add a cancellation reason field and report it.	High
F4	Customers	1,716 orders (34.32%) are pending. Their age is not shown.	Add an ageing view and a maximum waiting time.	High
F5	All pages	September looks partial: revenue ~₦0.6M, ~150 shipments, ~45 delivered, against ~₦2.6M and ~590 shipments in a normal month.	Confirm the last data date. Mark September as partial or exclude it from trends.	Medium
F6	Supplier	The order table shows Sum of UnitPrice as 66,043.45 on every OrderID, and the total shows the same figure. The measure appears to ignore the order filter.	Check the relationship between order items and orders and rebuild the measure.	High
F7	Customers, Shipments	There are 5,000 shipments for 5,000 orders although 1,626 orders are cancelled. Shipment value equals total revenue. Delivered shipments by month add up to ~1.8K, against 1,658 delivered orders.	Confirm whether cancelled orders have shipment records. Align delivery status with order status.	Medium
F8	Customers, Inventory	Ibadan places ~1,240 orders, but the location filters list only Abuja, Kano and Lagos.	Confirm how Ibadan orders are served.	Medium
F9	Inventory	380 low stock items are reported, but the threshold is not stated.	Document the rule behind the flag.	Medium
F10	Shipments	Shipments with status On Time still carry ~1.8K delay days, so the on-time rule seems to allow a grace period.	Document the on-time rule on the page.	Medium
F11	Inventory	The warehouse table has columns headed 0 to 7 with no label, and only 10 of 15 warehouses show without scrolling.	Label the column field and show all 15 warehouses.	Low
F12	Customers	The customer table is sorted by CustomerID, not by revenue.	Add a ranked top-customer view.	Low
F13	All pages	Not yet shown: product-level revenue and returns, return reasons, supplier performance score, inventory turnover, warehouse utilisation, average delivery time and a customer retention measure.	Add them, or list them as the next phase.	Medium
F14	Shipments	Visual titles such as “Delivery per DelayDays” are unclear to executives.	Rename to “Total Delay Days”.	Low

7. Part F: Strategic Recommendations

No.	Area	Based on	Recommendation
1	**Delivery optimisation**	36.28% on time; 3.57 days average delay (F2)	Agree a staged on-time target with management and review late shipments every week, warehouse by warehouse.
2	**Logistics optimisation**	Lagos holds ~54% of stock but ~25% of orders; Ibadan has no warehouse (F8)	Compare dispatch routes and stock placement with where customers are. Test moving stock closer to Kano, Abuja and Ibadan, and assess an Ibadan dispatch point.
3	**Warehouse efficiency**	Order load of 319 to 371 per warehouse; delay days per warehouse about equal	Review the shared process (picking, dispatch cut-off, carrier hand-over) in place of blaming one site. Add delay rate by warehouse to the dashboard.
4	**Inventory management**	380 low stock items; Food has the least stock (~0.305M) but the second-highest revenue (F9)	Document the low stock rule, set reorder points and clear the 380 items, starting with Food and fast-moving lines.
5	**Supplier improvement**	Revenue-only view; top suppliers earn ₦0.30M to ₦0.32M against a ₦144.45K average	Build a supplier scorecard covering on-time delivery, returns and revenue. Agree service levels with the lowest scorers.
6	**Operational cost reduction**	1,626 cancelled and 1,716 pending orders; 63.72% late (F3, F4)	Record the cost of cancellations and late deliveries, capture cancellation reasons, and escalate pending orders that pass a set waiting time.
7	**Return reduction**	9.46% return rate; Lagos ~250 returns	Add return reasons and a product-level return rate. Start with the products and location with the most returns and set a reduction target.
8	**Customer retention**	5.00 orders per customer; cancellations spread across customers	Add a retention measure and a ranked top-customer view. Contact top customers whose orders were cancelled or late.
9	**Performance monitoring**	Unfiltered revenue, the supplier table error and partial September (F1, F5, F6)	Make delivered-order revenue the headline KPI, fix F6, label partial months, name a dashboard owner and review the KPI cards weekly.
10	**Business growth**	Fashion at 37% of revenue; Kano highest on orders and revenue; peaks in April and August	Grow the Fashion range, plan stock and promotions ahead of April and August, and treat cancelled orders as the fastest source of recoverable revenue.

8. Summary

The SwiftLog dashboards bring orders, delivery, stock, returns and suppliers into one view, and the figures agree across the five pages. The main finding is that SwiftLog takes and ships orders at a steady rate but turns too few of them into on-time deliveries. Only 36.28% of shipments arrive on time, 33.16% of orders are delivered and 32.52% are cancelled. The ₦21.67M revenue figure is a booked total. If delivered orders carry the average value, delivered revenue is nearer ₦7.2M.

The delay pattern looks the same in every location, which points to a shared process issue rather than one weak site. Fourteen items were flagged, five of them rated High or Critical. The next steps are to fix the supplier table, confirm the September data, and add product, supplier and return measures so that the product and supplier questions can be answered in full.